New topic: <u>functions</u>.

<u>Math class:</u>	<u>English</u>	<u>C++</u>

$f: \mathbb{R} \to \mathbb{R}$	"f is a function from reals to reals"	`double f(double);`
		//Note: looks like
		//variable declaration...
		// but w/ (...)

$f(x) = x^2+1$	"f maps x to x^2+1"	`double f(double x)`
		`{`
		`  <u>return</u> x*x+1;`
		`}`

Note about <u>return</u>: the expression after will be the result of the function.
It also <u>ends the function's execution.</u>

Differences from math functions:

*Might not give the same result every time!

In math world, f(3) is ... f(3) it exists outside of time.

In C++, first call to f(3) might be 10, but the next call could give 11...

* Might have "side-effects", e.g. printing a message, writing to a file...

As such, functions with <u>no return value</u> might make sense!

```
void writefile(string frame, ...)
{
    ...
    return;
}
```

```
f(int x, int y, [universe])
```